

Day 1: Properties of Exponents (Variable Exponents)

- Simplify: $x^{n+3} \cdot x^{2n-1} =$ _____
- Simplify: $\frac{(3x^2y^{-3})^2}{9x^{-1}y^2} =$ _____
- SAT:** If $2^{3x} = 2^{12}$, what is the value of x ?
A. 4 B. 9 C. 15 D. 36
- If $a^{2x} \cdot a^3 = a^{15}$, what is the value of x ? _____

Day 2: Rational Exponents & Radicals

- SAT:** Which is equivalent to $\sqrt{x} \cdot \sqrt[3]{x}$?
A. $x^{1/6}$ B. $x^{2/5}$ C. $x^{5/6}$ D. $x^{6/5}$
- Simplify: $(x^{1/3})^{-6} =$ _____
- What is the value of $16^{-3/4}$?
A. -12 B. -8 C. $\frac{1}{8}$ D. 8
- Simplify completely: $\sqrt[3]{54x^8y^{12}} =$ _____

Day 3: Polynomial Operations

- Simplify: $(5x^3 - 2x^2 + 4x - 1) - (2x^3 + 3x^2 - x + 7) =$ _____
- Expand completely: $(x - 2)(x + 2)(x + 3)$

Show work:

Answer: _____

- SAT:** Which is equivalent to $(x + 1)^3$?
A. $x^3 + 1$ B. $x^3 + 3x^2 + 3x + 1$ C. $x^3 + 3x + 1$ D. $x^3 + x^2 + x + 1$

Day 4: Multiply Polynomials + Box Method Preview

- Expand: $(3x - 2)(2x^2 - x + 5) =$ _____
- Use the box to multiply $(x + 3)(x^2 - 2x + 4)$:

	x^2	$-2x$	$+4$
x			
$+3$			

Answer: _____

3. **SAT:** The expression $-x^2(-x)^4$ is equivalent to:

- A. x^6 B. $-x^6$ C. x^8 D. $-x^8$

Day 5: Division + Mixed Review

1. Simplify: $\frac{18x^5y^3 - 12x^3y^4 + 6x^2y^2}{6x^2y^2} =$ _____

2. **SAT:** The expression $\frac{(3y^3)(2y^2)^2}{(y^4)^3}$ is equivalent to:

- A. $\frac{12}{y^5}$ B. $12y^5$ C. $\frac{6}{y^5}$ D. $12y^{-5}$

3. **SAT:** If $\sqrt[3]{by} = b^5$, what is the value of y ?

- A. $\frac{5}{3}$ B. 5 C. 8 D. 15

4. Simplify: $(27x^6)^{2/3} =$ _____

Answer Key (remove before copying): **D1:** 1. x^{3n+2} 2. x^5/y^8 3. A 4. $x = 6$ **D2:** 1. C 2. $1/x^2$ 3. C 4. $3x^2y^4\sqrt[3]{2x^2}$ **D3:** 1. $3x^3 - 5x^2 + 5x - 8$ 2. $x^3 + 3x^2 - 4x - 12$ 3. B **D4:** 1. $6x^3 - 7x^2 + 17x - 10$ 2. $x^3 + x^2 - 2x + 12$ 3. B **D5:** 1. $3x^3y - 2xy^2 + 1$ 2. A 3. D 4. $9x^4$